

SUMMARY OF PRODUCT CHARACTERISTICS

1. NAME OF THE MEDICINAL PRODUCT

Iodopol, 37-7400 MBq, hard capsules

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

One capsule contains sodium iodide (^{131}I) 37 - 7400 MBq at time of calibration.

Iodine-131 is produced by fission of uranium-235 or by neutron bombardment of stable tellurium in a nuclear reactor. Iodine-131 has a half -life of 8.02 days. It decays by emission of gamma radiations of 365 keV (81.7%), 637 keV (7.2%) and 284 keV (6.1 %) and beta radiations of maximal energy of 606 keV to stable Xenon-131.

Excipients with known effect:

One hard capsule contains maximum 97 mg sodium per capsule.

One hard capsule contains quinolone yellow (E 104) 0.2% per capsule shell.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Hard capsule,

Medium orange hard gelatin capsule, approximately 18 mm long, containing white powder.

4. CLINICAL PARTICULARS

4.1. Therapeutic indications

Radioiodide thyroid therapy is indicated for:

- treatment of Graves' disease, toxic multinodular goitre or autonomous nodules.
- treatment of papillary and follicular thyroid carcinoma including metastatic disease.

Sodium Iodide (^{131}I) therapy is often combined with surgical intervention and with antithyroid medicinal products.

4.2. Posology and method of administration

The activity to be administered is a matter for clinical judgement. The therapeutic effect is only achieved after several months.

For the treatment of hyperthyroidism

The activity administered is usually in the range of 200 – 800 MBq but repeated treatment may be necessary. The dose required depends on the diagnosis, the size of the gland, thyroid uptake and iodine clearance. Patients should be rendered euthyroid medically whenever possible before giving radioiodine treatment for hyperthyroidism.

For the *Thyroid ablation and treatment of metastases*

The administered activities following total or subtotal thyroidectomy to ablate remaining thyroid tissue are in the range of 1850 – 3700 MBq. It depends on the remnant size and radioiodine uptake. In subsequent treatment of metastases, administered activity is in the range of 3700 – 11100 MBq.

$$\text{Paediatric dose (MBq)} = \frac{\text{Adult dose (MBq)} \times \text{child weight (kg)}}{70 \text{ kg}}$$

$$\text{Paediatric dose (MBq)} = \frac{\text{Adult dose (MBq)} \times \text{child surface (m}^2\text{)}}{1.73 \text{ m}^2}$$

Correction factors given for guidance are proposed below

Fraction of adult dose							
3 kg	=	0.1	22 kg	=	0.50	42 kg	= 0.78
4 kg	=	0.14	24 kg	=	0.53	44 kg	= 0.80
6 kg	=	0.19	26 kg	=	0.56	46 kg	= 0.82
8 kg	=	0.23	28 kg	=	0.58	48 kg	= 0.85
10 kg	=	0.27	30 kg	=	0.62	50 kg	= 0.88
12 kg	=	0.32	32 kg	=	0.65	52-54 kg	= 0.90
14 kg	=	0.36	34 kg	=	0.68	56-58 kg	= 0.92
16 kg	=	0.40	36 kg	=	0.71	60-62 kg	= 0.96
18 kg	=	0.44	38 kg	=	0.73	64-66 kg	= 0.98
20 kg	=	0.46	40 kg	=	0.76	68 kg	= 0.99

(Paediatric Task Group, European Association of Nuclear Medicines (EANM))

The capsule is administered orally together with a drink. It should be swallowed whole.

In patients with suspected gastrointestinal disease, great care should be taken when administering sodium [¹³¹I] iodide capsules. The capsules should be swallowed whole with sufficient fluid to ensure clear passing into the stomach and upper small intestine. Concomitant use of H₂ antagonists or proton pump inhibitors is advised.

After high doses used e.g. for the treatment of thyroid carcinoma, patients should be encouraged to increase oral fluids to have frequent bladder emptying to reduce bladder radiation.

4.3. Contraindications

- Hypersensitivity to the active substance or to any of the excipients
- Pregnancy.
- For diagnostic purposes in children under 10 years of age.
- Thyroid scanning except in the follow-up of malignant disease or when iodine-123 or technetium-99m is not available.
- Patients with dysphagia, oesophageal stricture, active gastritis, gastric erosions and peptic ulcer.
- Patients with suspected reduced gastrointestinal motility.

4.4. Special warnings and precautions for use

Potential for hypersensitivity or anaphylactic reactions

If hypersensitivity or anaphylactic reactions occur, the administration of the medicinal product must be discontinued immediately and intravenous treatment initiated, if necessary. To enable immediate action in emergencies, the necessary medicinal products and equipment such as endotracheal tube and ventilator must be immediately available.

Individual benefit/risk justification

For each patient, the radiation exposure must be justifiable by the likely benefit. The activity to be administered should in every case be as low as reasonably achievable to obtain the required therapeutic effect.

There is little evidence of an increased incidence of cancer, leukaemia or mutations in patients after treatment with radioiodine for benign thyroid diseases, despite its extensive use. In the treatment of malignant thyroid diseases, in a study conducted on patients with doses of sodium iodide (^{131}I) higher than 3,700 MBq a higher incidence of bladder cancer was reported. Another study reported a slight increase in leukaemia in patients receiving very high doses. Therefore total cumulative doses greater than 26,000 MBq are not recommended.

Gonadal function in males

The use of the sperm bank could be considered to compensate a potential reversible damage of gonadal function in males due to the high therapeutic dose of radioiodine, in the cases of patients with extensive disease.

Patients with renal impairment

Careful consideration of the benefit/risk ratio in these patients is required since an increased radiation exposure is possible. In these patients it may be necessary to adjust the posology.

Paediatric population

Careful consideration of the indication is required since the effective dose per MBq is higher than in adults (see section 11). When treating children and young adults, account must be taken of the greater sensitivity of child tissue and the greater life expectancy of such patients. The risks should be weighed against those of other possible treatments (see sections 4.2 and 11).

The radioiodine treatment of benign thyroid diseases of children and adolescents may be performed only in justified cases, especially in relapse after use of antithyroid medicinal products or in case

of serious adverse reactions to antithyroid medicinal products. There is no evidence of an increased incidence of cancer, leukemia or mutations in humans with respect to patients treated for benign thyroid disease with radioiodine, despite its extensive use.

Persons who have received radioiodine therapy of the thyroid as children and adolescents, should be reexamined once a year.

Hyponatraemia

Serious manifestations of hyponatraemia have been reported after sodium iodide (^{131}I) therapy in elderly patients who have undergone total thyroidectomy. Risk factors include older age, female sex, use of thiazide diuretics and hyponatraemia at the start of sodium iodide (^{131}I) therapy. Regular serum electrolytes measurements shall be considered for these patients.

Patient preparation

Patients should be encouraged to increase oral fluids and urged to void as often as possible to reduce bladder radiation, especially after high activities e.g. for the treatment of thyroid carcinoma. Patients with bladder voiding problems should be catheterised after administration of high activities of radioiodine.

To reduce colon radiation exposure, mild laxatives (but not stool softeners which do not stimulate the bowel) may be necessary in patients having less than one bowel movement a day.

To avoid sialadenitis that may occur after high dose radioiodine administration, the patient should be advised to take sweets or drinks containing citric acid (lemon juice, vitamin C) to stimulate saliva excretion before therapy. Other pharmacological protection measures may be used additionally.

Iodide overload from food or medicinal treatment should be investigated before administration of iodide (see section 4.5). A low iodine diet prior to therapy is recommended to enhance uptake into functioning thyroid tissue.

Thyroid replacement should be stopped prior to radioiodine administration for thyroid carcinoma to ensure adequate uptake. It is recommended to stop triiodothyronine treatment for a period of 14 days and to stop thyroxine treatment for a period of 4 weeks. They should be restarted two days after treatment. Carbimazole and propylthiouracil should be stopped 1 week prior to treatment of hyperthyroidism and restarted several days after treatment.

The radioiodine treatment of Graves' disease should be performed under concomitant treatment of corticosteroids, particularly when endocrine ophthalmopathy is present.

In patients with suspected gastrointestinal disease, great care should be taken when administering sodium iodide (^{131}I) capsules. Concomitant use of H₂-antagonists or proton pump inhibitors is advised.

After the procedure

Close contact with infants and pregnant women should be restricted for an appropriate period of time.

In case of vomiting, the risk of contamination has to be considered.

Patients treated with radioiodine should be followed-up at appropriate intervals.

Specific warnings

This medicinal product contains maximum 97 mg sodium per capsule, equivalent to 4.85% of the WHO recommended maximum daily intake of 2g sodium for an adult. To be taken into account by patients on a controlled sodium diet.

In patients with a known hypersensitivity for gelatine or its metabolites, sodium iodide (^{131}I) solution should be preferred for the radioiodine therapy.

This medicinal product contains azo colouring agent quinoline yellow (E 104). It may cause allergic reactions. It may have an adverse effect on activity and attention in children. Precautions with respect to environmental hazard, see section 6.6.

4.5. Interactions with other medicinal products and other forms of interaction

Many pharmacologically active substances interact with radioiodide. Various interaction mechanisms exist which can affect the protein binding, the pharmacokinetics or the dynamic effects of labelled iodide.

As a consequence, it should be considered that the thyroid uptake might be reduced. Therefore, a full drug history should be taken and relevant medicinal products are required to be withheld prior to the administration of sodium iodide (^{131}I).

For example, the treatment with the following substances should be discontinued:

Active substances	Withdrawal period before administration of sodium iodide (^{131}I)
Antithyroid medicinal products (e.g. carbimazole, methimazole, propyluracil), perchlorate	1 week before starting treatment till several days after
Salicylates, corticosteroids, sodium nitroprusside, sodium sulfobromophthalein, anticoagulants, antihistamines, antiparasitics, penicillins, sulphonamides, tolbutamide, thiopental	1 week
Phenylbutazone	1 - 2 weeks
Containing iodine expectorants and vitamins	approximately 2 weeks
Thyroid hormone preparations	Triiodothyronine 2 weeks thyroxine 6 weeks
Benzodiazepines, lithium	approximately 4 weeks
Amiodarone*	3 – 6 months
Containing iodine preparations for topical use	1 - 9 months
Water-soluble iodine-containing contrast media	6 to 8 weeks
Lipo-soluble iodine-containing contrast media	up to 6 months

* Due to the long half-life of amiodarone, iodine uptake in the thyroid tissue can be decreased for several months.

4.6. Fertility, pregnancy and lactation

Women of childbearing potential

When an administration of radiopharmaceuticals to a woman of childbearing potential is intended, it is important to determine whether or not she is pregnant. Any woman who has missed a period should be assumed to be pregnant until proven otherwise. If in doubt about her

potential pregnancy (if the woman has missed a period, if the period is very irregular, etc.), alternative techniques not using ionising radiation (if there are any) should be offered to the patient. Women receiving sodium iodide (^{131}I) should be advised not to become pregnant within 6-12 months after administration.

Contraception in males and females

Contraception for 6 months (for patients with benign thyroid conditions) or 12 months (for patients with thyroid cancer) is recommended for both sexes after therapeutic administration of sodium iodide (^{131}I). Men should not father a child for a time period of 6 months after radioiodine treatment to allow the replacement of irradiated by non-irradiated spermatozoa. Sperm banking should be considered for men who have extensive disease and therefore may need high sodium iodide (^{131}I) therapeutic doses.

Pregnancy

The use of sodium iodide (^{131}I) is contraindicated during established or suspected pregnancy or when pregnancy has not been excluded because transplacental passage of sodium iodide (^{131}I) can cause severe and possibly irreversible hypothyroidism in neonates (the absorbed dose to the uterus for this medicinal product is likely to be in the range 11-511 mGy, and the foetal thyroid gland avidly concentrates iodine during the second and third trimesters) (see section 4.3). If a differentiated thyroid carcinoma is diagnosed during pregnancy, sodium iodide (^{131}I) treatment should be postponed until after the childbirth.

Breast-feeding

Before administering radiopharmaceuticals to a mother who is breast-feeding, consideration should be given to the possibility of delaying the administration of radionuclide until the mother has ceased breast-feeding, and what is the most appropriate choice of radiopharmaceuticals, bearing in mind the secretion of activity in breast milk.

If the administration is considered necessary, breast-feeding must be discontinued at least 8 weeks before sodium iodide (^{131}I) administration and should not be resumed thereafter (see section 4.3). For radioprotection reasons following therapeutic doses, it is recommended to avoid close contact between mother and infants for at least one week.

Fertility

After radioiodine therapy of thyroid carcinoma, a dose dependent impairment of fertility may occur in men and women. Depending on the activity dose, a reversible impairment of the spermatogenesis could occur in doses above 1,850 MBq. Clinical relevant effects including oligospermia and azospermia and elevated serum FSH levels have been described after administration greater than 3,700 MBq.

4.7. Effects on ability to drive and use machines

Sodium iodide (^{131}I) has no or negligible influence on the ability to drive or use machines.

4.8. Undesirable effects

Summary of the safety profile

The frequencies of reported adverse reactions were derived from the medical literature. The safety profile of sodium iodide (^{131}I) differs widely according to the doses administered, while the doses to be administered are dependent on the type of treatment (i.e. treatment of benign or malignant disease). Moreover, the safety profile depends on the cumulative doses administered and the dosing intervals which are used. Therefore, the reported adverse reactions were grouped by their occurrence in treatment of benign or malignant disease.

Frequently occurring adverse reactions are: hypothyroidism, transient hyperthyroidism, salivary and lacrimal gland disorders, and radiation local effects. In cancer treatment additionally gastrointestinal adverse reactions and bone marrow suppression may frequently occur.

The following tables include reported adverse reactions sorted by system organ classes.

Symptoms, which are rather secondary to a group-syndrome (e.g. sicca syndrome) are subsumed in parenthesis behind the respective syndrome.

Tabulated list of adverse reactions

The following table presents how the frequencies are reflected in this section:

Very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1,000$ to $< 1/100$), rare ($\geq 1/10,000$ to $< 1/1,000$), very rare ($< 1/10,000$) and not known (frequency cannot be estimated from the available data).

Adverse reactions after treatment of benign disease

<i>System Organ Class</i>	<i>Adverse reaction</i>	<i>Frequency</i>
Immune system disorders	Anaphylactoid reaction	Not known
Endocrine disorders	Permanent hypothyroidism, hypothyroidism	Very common
	Transient hyperthyroidism	Common
	Thyreotoxic crisis, thyroiditis, hypoparathyroidism (blood calcium decreased, tetany)	Not known
Eye disorders	Endocrine ophthalmopathy (in Graves' disease)	Very common
	Sicca syndrome	Not known
Respiratory thoracic and mediastinal disorders	Vocal cord paralysis	Very rare
Gastrointestinal Disorders	Sialoadenitis	Common
Hepatobiliary disorders	Hepatic function abnormal	Frequency not know
Skin and subcutaneous tissue disorders	Iodide induced acne	Not known
Congenital, familial and genetic disorders	Congenital hypothyroidism	Not known
General disorders and administration site conditions	Local swelling	Not known

Adverse reactions after treatment of malignant disease

<i>System Organ Class</i>	<i>Adverse reaction</i>	<i>Frequency</i>
Neoplasms benign, malignant and unspecified (including cysts and polyps)	Leukaemia	Uncommon
	Solid cancers, bladder cancer, colon cancer, gastric cancer, breast cancer	Not known
Blood and lymphatic system disorders	Erythropenia, bone marrow failure	Very common
	Leukopenia, thrombocytopenia	Common
	Aplastic anemia, permanent or severe bone marrow suppression	Not known
Immune system disorders	Anaphylactoid reaction	Not known
Endocrine disorders	Thyreotoxic crisis, transient hyperthyroidism	Rare
	Thyroiditis (transient leucocytosis), hypoparathyroidism (blood calcium decreased, tetany), hypothyroidism, hyperparathyroidism	Not known
Nervous system disorders	Parosmia, anosmia	Very common
	Brain oedema	Not known
Eye disorders	Sicca syndrome (conjunctivitis, dry eyes, nasal dryness)	Very common
	Nasolacrimal duct obstruction (lacrimation increased)	Common
Respiratory, thoracic and mediastinal disorders	Dyspnoea	Common
	Throat constriction*, pulmonary fibrosis, respiratory distress, obstructive airways disorder, pneumonia, tracheitis, vocal cord dysfunction (vocal cord paralysis, dysphonia, hoarseness), oropharyngeal pain, stridor	Not known
Gastrointestinal disorders	Sialoadenitis (dry mouth, salivary gland pain, salivary gland enlargement, dental caries, tooth loss), radiation sickness syndrome, nausea, ageusia, anosmia, dysgeusia, decreased appetite	Very common
	Vomiting	Common
	Gastritis, dysphagia	Not known
Hepatobiliary disorders	Hepatic function abnormal	Frequency not know
Renal and urinary disorders	Radiation cystitis	Not known
Reproductive system and breast disorders	Ovarian failure, menstrual disorder	Very common
	Azoospermia, oligospermia, decreased fertility male	Not known
Congenital, familial and genetic disorders	Congenital hypothyroidism	Not known
Metabolism and nutrition disorders	Hyponatraemia	Not known
General disorders and administration site conditions	Flu-like illness, headache, fatigue, neck pain	Very common
	Local swelling	Common

*: especially in existing tracheal stenosis

Description of selected undesirable effects

General advice

Exposure to ionising radiation is linked with cancer induction and a potential for development of hereditary defects. The radiation dose resulting from therapeutic exposure may result in higher incidence of cancer and mutations. In all cases it is necessary to ensure that the risks of the radiation are less than those of the disease itself. The effective dose after therapeutic doses of sodium iodide (^{131}I) is 3,108 mSv when the maximal recommended activity of 11,100 MBq (with thyroid uptake 0%).

Thyroid and parathyroid gland disorders

Hypothyroidism may occur, depending on the dose, as a delayed result of treatment for hyperthyroidism with radioiodine.

In the treatment of malignant disease, hypothyroidism is often reported as an adverse reaction; however the treatment of malignant diseases with radioiodine generally follows thyroidectomy.

The destruction of thyroid follicles caused by the radiation exposure of sodium iodide (^{131}I) may lead to exacerbation of an already existing hyperthyroidism within 2 – 10 days or may cause a thyrotoxic crisis. Occasionally, an immune hyperthyroidism may appear after initial normalisation (latency period is 2 – 10 months). After 1 - 3 days of administration of high dose radioiodine, the patient may experience transient inflammatory thyroiditis and tracheitis, with a possibility of severe tracheal constriction, especially where there is existing tracheal stenosis.

In rare cases, a temporary hyperthyroidism could be observed even after treatment of a functional thyroid carcinoma.

Cases of transient hypoparathyroidism have been observed after radioiodine administration which should be appropriately monitored and treated with replacement therapy.

Late consequences

Dose dependent hypothyroidism may occur as a delayed result of radioiodine treatment of hyperthyroidism. This hypothyroidism may manifest itself weeks or years after the treatment, and monitoring of thyroid function and appropriate hormone replacement therapy are required. Hypothyroidism does not generally appear until 6 - 12 weeks after radioiodine administration.

Eye disorders

Endocrine ophthalmopathy may progress or new ophthalmopathy may occur after radioiodine therapy of hyperthyroidism or Graves' disease. Radioiodine treatment of Graves' disease should be associated with corticosteroids.

Local irradiation effects

Dysfunction and paralysis of vocal cord have been reported after administration of sodium iodide (^{131}I), however, in some cases it cannot be decided whether the dysfunction of the vocal cords was caused by radiation or by surgical treatment.

High tissue uptake of radioiodine can be associated with local pain, discomfort and local oedema e.g. in case of radioiodine treatment of the remnant thyroid gland, a diffuse and severe soft tissue pain may occur in the head and neck region.

Radiation induced pneumonia and pulmonary fibrosis have been observed in patients with diffuse pulmonary metastases from differentiated thyroid carcinoma, due to destruction of metastatic tissue. This occurs mainly after high dose radioiodine therapy.

In the treatment of metastasing thyroid carcinomas with central nervous system (CNS) involvement, the possibility of local cerebral oedema and/or aggravation of existing cerebral oedema should also be considered.

Gastrointestinal disorders

High levels of radioactivity may also lead to gastrointestinal disturbance, usually within the first hours or days after administration. For prevention of gastrointestinal disorders see section 4.4.

Salivary and lacrimal gland disorders

Sialoadenitis may occur, with swelling and pain in the salivary glands, partial loss of taste and dry mouth. Sialoadenitis is usually reversible spontaneously or with anti-inflammatory treatment but cases of dosedependent persistent ageusia and dry mouth have occasionally been described. The lack of saliva may lead to infections, e.g. caries and this may result in loss of teeth. For prevention of salivary disorders see section 4.4.

Malfunction of the salivary and/or lacrimal glands with resulting sicca syndrome may also appear with a delay of several months and up to two years after radioiodine therapy. Although sicca syndrome is a transient effect in most cases, the symptom may persist for years in some patients.

Bone marrow depression

As a late consequence, reversible bone marrow depression may develop, presenting with isolated thrombocytopenia or erythrocytopenia which may be fatal. Bone marrow depression is more likely to occur after one single administration of more than 5,000 MBq, or after repeat administration in intervals below 6 months.

Secondary malignancies

After higher activities, typically those used in the treatment of thyroid malignancies, an increased incidence of leukaemia has been observed. There is evidence of an increased frequency of solid cancers induced by administration of high activities (above 7.4 GBq).

Paediatric population

The type of undesirable effects expected in children are identical to the one in adults. Based on greater radiation sensitivity of child tissues (see section 11.) and the greater life expectancy frequency and severity may be different.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system. Adverse reactions may be reported to marketing authorization holder.

4.9. Overdose

This product must be used by authorized personnel in hospital setting. The risk of overdose is therefore theoretical.

In the event of administration of a radiation overdose, the absorbed dose to the patient should be reduced where possible by increasing the elimination of the radionuclide from the body by frequent micturition and by forced diuresis and frequent bladder voiding. Additionally, the blockade of the thyroid gland should be recommended (e.g. with potassium perchlorate) in order to reduce the radiation exposure of the thyroid gland. To reduce the uptake of iodine (^{131}I), emetics can be given.

5. PHARMACOLOGICAL PROPERTIES

5.1. Pharmacodynamic properties

Pharmacotherapeutic group: therapeutic radiopharmaceuticals, Iodine (^{131}I) compounds

ATC code: V 10X A01

The pharmacological active substance is iodine (^{131}I) in the form of sodium iodide that is taken up by the thyroid. The physical decay takes place essentially in the thyroid gland where iodide-131 has a long residence time, delivering a selective irradiation to this organ.

In the amount used for therapeutic indications, no pharmacodynamic effects of sodium iodide (^{131}I) are to be expected.

More than 90% of the radiation effects result from emitted β radiation which has a mean range of 0.5 mm. The β radiation dose will dependently decrease cell function and cell division leading to cell destruction. The short range and almost absence of uptake of sodium iodide (^{131}I) outside the thyroid lead to a negligible irradiation exposure outside the thyroid gland.

5.2. Pharmacokinetic properties

Absorption

After oral administration sodium iodide (^{131}I) is absorbed rapidly from the upper gastrointestinal tract (90% in 60 minutes). The absorption is influenced by gastric emptying. It is increased by hyperthyroidism and decreased by hypothyroidism.

Studies on the serum activities levels showed that after a fast increase, over 10 to 20 minutes, an equilibrium is reached after about 40 minutes. After oral administration of a sodium iodide (^{131}I) solution an equilibrium is reached at the same time.

Organ distribution and uptake

The pharmacokinetics follows that of unlabelled iodide. After entering the blood stream it is distributed in the extra thyroidal compartment. From here it is predominantly taken up by the thyroid that extracts approximately 20% of the iodide in one pass or excreted renally. The iodide uptake in the thyroid reaches a maximum after 24 - 48 hours, 50% of the maximum peak is reached after 5 hours. The uptake is influenced by several factors: patient age, thyroid gland volume, renal clearance, plasmatonic concentration of iodide and other drugs (see section 4.5). The iodide clearance

by the thyroid gland is usually 5 - 50 mL/min. In case of iodine deficiency the clearance is increased to 100 mL/min and in case of hyperthyroidism can be up to 1,000 mL/min. In case of iodide overload the clearance can decrease to 2 – 5 mL/min. Iodide also accumulates in the kidneys. Small amounts of sodium iodide (^{131}I) are taken up by salivary glands, gastric mucosa and they would also be localised in breast milk, the placenta and choroid plexus.

The iodide fixed by the thyroid enters the known metabolic path of thyroid hormones and is incorporated in the organic substances entering in the synthesis of thyroid hormones.

Biotransformation

The iodide that has been taken up by the thyroid follows the known metabolism of the thyroid hormones and is incorporated in the organic compounds from which the thyroid hormones are synthesised.

Elimination

Urinary excretion is 37-75%, faecal excretion is about 10% with almost negligible excretion in sweat. Urinary excretion is characterised by the renal clearance, which constitutes about 3% of the renal flow and is relatively constant from one person to another. The clearance is lower in hypothyroidism and in impaired renal function and higher in hyperthyroidism. In euthyroidic patients with normal renal function 50 – 75 % of the administered activity is excreted in urine within 48 hours.

Half-life

The effective half-life of radioiodine in plasma is about 12 hours in blood plasma and about 6 days in the thyroid gland. Thus after administration of sodium iodide (^{131}I) about 40% of the activity has an effective half-life of 6 hours and the remaining 60% of 8 days.

Renal impairment

Patients with renal impairment may have a decrease in the radioiodine clearance, resulting in increased radiation exposure of sodium iodide (^{131}I) administered. One study showed, for example, that patients with impaired renal function undergoing continuous ambulatory peritoneal dialysis (CAPD) have a clearance of radioiodine 5 times lower than patients with normal kidney function.

5.3. Preclinical safety data

Because of the small quantities of administered substance compared with the normal intake of iodine with food (40-500 µg/day) no acute toxicity is expected or observed.

There are no data available on the toxicity of repeated doses of sodium iodide nor on its effects on reproduction in animals or its mutagenic or carcinogenic potential.

6. PHARMACEUTICAL PARTICULARS

6.1. List of excipients

Capsule contents:

Sodium carbonate

Sodium hydrogen carbonate

Sodium hydroxide

Disodium phosphate dihydrate

Sodium thiosulfate

Gelatin capsule shell composition:

Quinoline yellow (E 104)

Erythrosine (E 127)

Titanium dioxide (E 171)

Gelatin

Source of gelatin capsule: Bovine

6.2. Incompatibilities

Not applicable.

6.3. Shelf life

21 days from the manufacturing date

6.4. Special precautions for storage

Store below 30°C.

Store in the original package to avoid radiation exposure from the product.

Storage of radiopharmaceuticals should be in accordance with national regulations on radioactive materials.

Do not freeze. Open the tin can under a ventilated hood because Xe-131 may have formed during storage.

6.5. Nature and contents of container

The polypropylene vial placed in a shielding lead container, closed with a lead stopper which contains polypropylene insert with iodine absorber. The package contains a single capsule. Each package is accompanied by a separate polypropylene applicator for capsule administration.

6.6. Special precautions for disposal and other handling

General warning

Radiopharmaceuticals should be received, used and administered only by authorized persons in designated clinical settings. Their receipt, storage, use, transfer and disposal are subject to the regulations and/or appropriate licences of the competent official organization.

Radiopharmaceuticals should be prepared in a manner which satisfies both radiation safety and pharmaceutical quality requirements.

Precautions to be taken before handling or administration of the medicinal product

The administration of sodium iodide (^{131}I) for therapy is likely to result in a relatively high radiation dose to most patients and may result in significant environmental hazard and creates risks for other persons from external radiation or contamination from spill of urine, vomiting, etc. This may be of concern to the immediate family of those individuals undergoing treatment or the general public depending on the level of activity administered.

Suitable precautions in accordance with national regulations should therefore be taken.

Administration procedures should be carried out in a way to minimize risk of contamination of the medicinal product and irradiation of the operators. Adequate shielding is mandatory.

When opening the container personnel should be aware that free radioactivity may be registered on monitors. This activity is due to Xe-131m which is formed for 1.17 % in the decay of I-131. Though visible on monitors this does not pose a relevant risk for personnel.

The effective dose rate by inhalation of the Xe-131m formed is 0.1% of the dose rate at 1 m from a lead- shielded capsule.

Precautions and activity data

1.3% of iodine (^{131}I) decays via xenon (^{131}mXe) (half-life 12 days) and a small amount of xenon (^{131}mXe) activity may be present in the packaging as a result of diffusion. It is therefore recommended that the transport container be opened in a ventilated enclosure and that, after removal of the capsule, the packaging materials are allowed to stand overnight before disposal to permit the release of absorbed xenon (^{131}mXe). In addition, there can be limited leakage of volatile iodine-131 activity from the capsule. The container incorporates a small disc of charcoal in the lid which serves to absorb the iodine that escapes from the capsule. The charcoal disc may become contaminated with up to 1.3MBq (35 μCi) of iodine-131. As a consequence of the charcoal disc, only very small amounts of iodine-131 (typically less than 1.85kBq (50nCi)) may be present in the packaging.

The activity of a capsule at 12h00 GMT from calibration date can be calculated from the table 1.

Table 1

Day	Coefficient	Day	Coefficient
-6	1.677	5	0.650
-5	1.539	6	0.596
-4	1.412	7	0.547
-3	1.295	8	0.502
-2	1.188	9	0.460
-1	1.090	10	0.422
0	1.000	11	0.387
1	0.917	12	0.355
2	0.842	13	0.326
3	0.772	14	0.299
4	0.708		

Any unused product or waste material should be disposed of in accordance with local requirements.

7. PRODUCT REGISTRATION HOLDER

MEDI RADIO MALAYSIA Sdn Bhd,
Plaza Mont Kiara,
E-13A-9, No 2. Jalan Kiara,
Mont Kiara, 50480 Kuala Lumpur, Malaysia

Manufacturer:

Narodowe Centrum Badan Jadrowych
Ul. Andrzej Soltana 7
05-400 Otwock, Poland

8. PRODUCT REGISTRATION NUMBER

MALxxxxxxxxAZ

9. DATE OF FIRST AUTHORISATION/RENEWAL OF THE AUTHORISATION

10. DATE OF REVISION OF THE TEXT

24-Oct-2023

11. DOSIMETRY

The data listed below are from ICRP (International Commission on Radiological Protection. Radiation Dose to Patients from Radiopharmaceuticals) publication 128.

The biokinetic model is described as a compartment model including inorganic iodide as well as organically bound iodine released to the body tissues following discharge from the thyroid. The ICRP model refers to oral administration.

As part of the risk-benefit assessment it is advised that the effective dose and likely radiation doses to individual target organ(s) are calculated prior to administration. The activity might then be adjusted according to thyroid volume, biological half-life and the “re-cycling” factor which takes into account the physiological status of the patient (including iodine depletion) and the underlying pathology.

Doses to the following target organs can be used:

Autonomous	Target organ dose 300 – 400 Gy
Toxic multinodular goitre	Target organ dose 150 – 200 Gy
Graves’ disease	Target organ dose 200 Gy
Large euthyroid goitre	Target organ dose 100 – 150 Gy

The radiation exposure mainly affects the thyroid. The radiation exposure of the other organs is in the range of thousandths lower than that of the thyroid. It depends on the dietary intake of iodine (the uptake of radioiodine is increased up to 90% in iodine deficient areas and it is decreased to 5%

in iodine rich areas). It further depends on the thyroid function (eu-, hyper-, or hypothyroidism) and on the presence of iodine accumulating tissues in the body (e.g. the situation after excision of the thyroid, the presence of iodine accumulating metastases and on thyroid blockade) The radiation exposure of all other organs is correspondingly higher or lower, depending on the degree of accumulation in the thyroid.

Thyroid blocked, uptake 0 %, oral administration

Organ	Absorbed dose per unit activity administered (mGy/MBq)				
	Adult	15 years	10 years	5 years	1 year
Adrenals	0.044	0.054	0.086	0.14	0.25
Bone surfaces	0.030	0.037	0.059	0.092	0.18
Brain	0.021	0.026	0.043	0.071	0.14
Breast	0.020	0.025	0.042	0.069	0.13
Gallbladder Wall	0.037	0.048	0.085	0.13	0.21
GI-tract					
Stomach wall	0.87	1.1	1.6	2.8	5.9
Small intest wall	0.035	0.044	0.070	0.11	0.19
Colon wall	0.14	0.18	0.30	0.50	0.92
(ULI wall	0.12	0.15	0.25	0.42	0.75)
(LLI wall	0.17	0.22	0.37	0.61	1.2)
Heart wall	0.062	0.080	0.13	0.20	0.37
Kidneys	0.062	0.080	0.13	0.20	0.37
Liver	0.050	0.065	0.10	0.16	0.30
Lungs	0.053	0.068	0.11	0.18	0.36
Muscles	0.026	0.032	0.051	0.080	0.15
Oesophagus	0.024	0.030	0.049	0.079	0.15
Ovaries	0.038	0.049	0.076	0.11	0.20
Pancreas	0.060	0.073	0.11	0.16	0.28
Red marrow	0.031	0.038	0.061	0.095	0.18
Salivary glands	0.27	0.33	0.44	0.59	0.86
Skin	0.019	0.023	0.038	0.062	0.12
Spleen	0.064	0.077	0.12	0.19	0.34
Testes	0.025	0.033	0.055	0.084	0.15
Thymus	0.024	0.030	0.049	0.079	0.15
Thyroid	2.2	3.6	5.6	0.13	0.25
Urinary bladder wall	0.54	0.71	1.1	1.4	1.8
Uterus	0.045	0.037	0.062	0.10	0.18
Remaining organs	0.029	0.037	0.060	0.10	0.18
Effective dose (mSv/MBq)	0.28	0.40	0.54	1.1	2.0

Thyroid low uptake, oral administration

Organ	Absorbed dose per unit activity administered (mGy/MBq)				
	Adult	15 years	10 years	5 years	1 year
Adrenals	0.051	0.067	0.12	0.20	0.44
Bone surfaces	0.089	0.10	0.14	0.22	0.40

Brain	0.093	0.10	0.13	0.18	0.30
Breast	0.038	0.050	0.10	0.17	0.32
Gallbladder wall	0.043	0.057	0.1	0.18	0.36
GI-tract					
Stomach wall	0.77	1.0	1.5	2.5	5.3
Small intestine	0.033	0.043	0.073	0.11	0.22
(ULI wall	0.12	0.15	0.27	0.49	1.0)
(LLI wall	0.17	0.22	0.39	0.71	1.6)
Heart wall	0.089	0.12	0.21	0.36	0.77
Kidneys	0.27	0.34	0.50	0.84	1.8
Liver	0.093	0.14	0.24	0.46	1.2
Lungs	0.10	0.13	0.22	0.38	0.79
Musclers	0.084	0.11	0.17	0.27	0.48
Oesophagus	0.10	0.15	0.30	0.58	1.1
Ovaries	0.037	0.049	0.080	0.13	0.28
Pancreas	0.064	0.080	0.13	0.21	0.41
Red marrow	0.072	0.086	0.12	0.19	0.37
Salivary glands	0.22	0.27	0.36	0.49	0.72
Skin	0.043	0.053	0.080	0.12	0.25
Spleen	0.069	0.089	0.15	0.26	0.55
Testes	0.024	0.032	0.056	0.095	0.20
Thymus	0.10	0.15	0.30	0.59	1.1
Thyroid	280	450	670	1400	2300
Urinary bladder wall	0.45	0.58	0.89	1.2	1.6
Uterus	0.042	0.054	0.090	0.15	0.28
Remaining organs	0.048	0.1111	0.17	0.25	0.44
Effective Dose (mSv/MBq)	14	23	34	71	110

Thyroid medium uptake, oral administration

Organ	Absorbed dose per unit activity administered (mGy/MBq)				
	Adult	15 years	10 years	5 years	1 year
Adrenals	0.055	0.047	0.13	0.24	0.55
Bone surfaces	0.12	0.14	0.19	0.30	0.52
Brain	0.13	0.14	0.18	0.24	0.39
Breast	0.048	0.063	0.13	0.23	0.43
Gallbladder wall	0.046	0.063	0.12	0.21	0.45
GI-tract					
Stomach wall	0.71	0.95	1.4	2.4	5.0
Small intestine	0.032	0.043	0.075	0.11	0.24
wall					
Colon wall	0.14	0.18	0.34	0.63	1.4
(ULI wall	0.12	0.15	0.28	0.53	1.2)
(LLI wall	0.17	0.22	0.40	0.76	1.8)
Heart wall	0.10	0.14	0.25	0.45	1.0
Kidneys	0.27	0.34	0.53	0.93	2.1
Liver	0.12	0.18	0.31	0.62	1.7
Lungs	0.13	0.16	0.28	0.50	1.0
Muscles	0.12	0.15	0.24	0.38	0.66
Oesophagus	0.14	0.22	0.45	0.87	1.7

Ovaries	0.036	0.049	0.082	0.15	0.33
Pancreas	0.066	0.084	0.14	0.24	0.49
Red marrow	0.095	0.11	0.15	0.24	0.48
Salivary glands	0.19	0.24	0.32	0.43	0.64
Skin	0.057	0.070	0.10	0.16	0.33
Spleen	0.023	0.032	0.056	0.10	0.23
Testes	0.023	0.032	0.056	1.0	2.3
Thymus	0.14	0.22	0.45	0.87	1.7
Thyroid	430	690	1000	2200	3600
Urinary bladder wall	0.39	0.51	0.79	1.1	1.5
Uterus	0.040	0.053	0.089	0.15	0.32
Remaining organs	0.11	0.15	0.23	0.33	0.58
Effective Dose (mSv/MBq)	22	35	53	110	180

Thyroid high uptake, oral administration

Organ	Absorbed dose per unit activity administered (mGy/MBq)				
	Adult	15 years	10 years	5 years	1 year
Adrenals	0.059	0.082	0.15	0.28	0.66
Bone surfaces	0.16	0.18	0.24	0.37	0.65
Brain	0.17	0.18	0.24	0.37	0.65
Breast	0.058	0.077	0.17	0.28	0.54
Gallbladder	0.049	0.068	0.13	0.24	0.54
GI-tract					
Stomach wall	0.66	0.88	1.3	2.2	4.7
Small intestine wall	0.032	0.043	0.077	0.12	0.26
Colon wall	0.14	0.19	0.35	0.68	0.16
(ULI wall	0.12	0.16	0.30	0.58	1.4)
(LLI wall	0.16	0.22	0.42	0.81	2.0
Heart wall	0.12	0.16	0.30	0.58	1.4
Kidneys	0.27	0.35	0.55	1.0	2.4
Liver	0.14	0.22	0.39	0.79	2.2
Lungs	0.15	0.20	0.35	0.61	1.3
Muscles	0.15	0.19	0.31	0.49	0.86
Oesophagus	0.19	0.28	0.59	1.2	2.3
Ovaries	0.035	0.049	0.084	0.16	0.37
Pancreas	0.068	0.088	0.15	0.27	0.57
Red marrow	0.12	0.14	0.19	0.29	0.59
Salivary glands	0.16	0.20	0.27	0.37	0.55
Skin	0.071	0.087	0.13	0.19	0.41
Spleen	0.075	0.10	0.18	0.33	0.80
Testes	0.22	0.031	0.057	0.11	0.27
Thymus	0.19	0.28	0.59	1.2	2.3
Thyroid	580	940	1400	3000	4900
Urinary bladder wall	0.34	0.44	0.68	0.95	1.3
Uterus	0.038	0.051	0.089	0.16	0.36
Remaining organs	0.15	0.19	0.29	0.42	0.74

Effective Dose (mSv/MBq)	29	47	71	150	250
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12. INSTRUCTIONS FOR PREPARATION OF RADIOPHARMACEUTICALS

The capsules are ready for use. Determine the activity before use.

Instructions for opening the container with the radioactive product using the applicator:

1. Check the radioactivity and calibration date placed on the outer package
2. Tear off the upper cover of the shipping container (metal tin).
3. Remove the upper styrofoam inlay.
4. Take the capsule shielding container out.
5. Tear the paper-foil mouthpiece wrapping and take out the mouthpiece
6. Open the capsule shielding container. To do this, hold the bottom part of the container and pull the upper part upwards. The vial containing the capsule should remain in the shielding container.
7. Connect the mouthpiece to the vial. To do this, screw in the mouthpiece into the vial containing the capsule.
8. During the administration of the capsule it is recommended to keep the vial containing the capsule in the shielding container. The patient holding the shielding container in his hand takes the mouthpiece in his mouth and then tilts it to get the capsule from the vial through the mouthpiece. When required, it is possible to administer a capsule without using the shielding container. The patient grasps the mouthpiece, takes the capsule vial out from the shielding container, takes the mouthpiece in his mouth and then tilts it to get the capsule from the vial through the mouthpiece.
9. After the administration of the capsule, the mouthpiece and the vial should be disposed of. The shielding container should be returned to the manufacturer.
10. To disconnect the mouthpiece from the vial, put the vial with the mouthpiece in the shielding container and then, holding the container with your hand, screw off the mouthpiece in order to disconnect it.
11. In order to measure the capsule activity, take the mouthpiece fixed to the capsule vial with the gripping device of the dose calibrator and load in the dose calibrator. When the measurement is finished remove the mouthpiece fixed to the capsule vial and place it back in the shielding container. When transferring the capsule to another room is necessary, the mouthpiece should be disconnected from the vial according to above instruction. After disconnecting the mouthpiece, cover the shielding container with a lid.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

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